

Solution Requirements

Date	29 June 2026
Team ID	xxxxxx
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The current system does not require user authentication. Future versions may include secure login for farmers and administrators.	Low
2	Authorization levels	All users can access the concept analysis system. Future administrator access may allow management of reference concepts, AI models, and user reports.	Low

3	External interfaces	The system interacts with OpenAI Whisper for speech transcription, Sentence-BERT for semantic similarity evaluation, Librosa for audio analysis, and ReportLab for PDF report generation.	High
4	Transactions processing	The system accepts an audio file, transcribes speech, analyzes semantic similarity and speech features, computes an understanding score, and displays the evaluation results.	High
5	Reporting	Generate and display transcripts, semantic similarity scores, speech analysis metrics, final evaluation scores, and downloadable PDF reports.	Medium
6	Business rules, etc.	Accept only supported audio formats (WAV, MP3, M4A, OGG). Validate uploaded files before analysis and calculate the final understanding score using semantic similarity and speech quality metrics.	High
7	Compliance to laws or regulations	Follow secure coding practices and ensure uploaded audio files are processed securely without storing personally identifiable information.	Medium
8	Other	Support browser-based access through a responsive Streamlit interface and provide an interactive user experience.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Analyze uploaded audio and generate evaluation results efficiently.	Complete analysis within 10–20 seconds for a typical 1–2 minute audio file.
2	Scalability	Support multiple users analyzing audio simultaneously with minimal performance degradation.	Handle multiple concurrent requests while maintaining stable response times.
3	Security & Data Privacy	Validate uploaded audio files and prevent malicious or unsupported file uploads. Protect user data during processing.	100% file validation; no permanent storage of uploaded audio without user consent.
4	Reliability & Availability	Ensure consistent prediction results and uninterrupted system operation.	99%+ availability during deployment
5	Usability & Accessibility	Provide a clean, intuitive, and responsive interface for uploading audio, viewing results, and downloading reports..	Responsive Streamlit interface with clear navigation and easy-to-understand results.
6	<i>Other</i>	Maintain a modular architecture so AI models and evaluation components can be updated independently.	Separate modules (app.py, speech_to_text.py, semantic_eval.py, audio_utils.py, scoring_engine.py, report_generator.py) with reusable code.